

A Geospatial Comparative Analysis of Small Area Estimation Models for Antenatal Care Coverage in Kenya

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1 Introduction

Ensuring adequate access to antenatal care (ANC) is a cornerstone of maternal and child health, yet achieving equitable coverage remains a significant challenge in developing nations like Kenya. While national-level data provide a broad overview of health trends, they often mask severe sub-national disparities. In Kenya, significant gaps exist between urban hubs like Nairobi and Arid and Semi-Arid Lands (ASAL) such as Mandera and Turkana.

The primary challenge in addressing these disparities lies in the unreliability of traditional survey data at local levels. Direct estimates from the Kenya Demographic and Health Survey (KDHS) frequently suffer from high variability and wide confidence intervals in sparsely populated areas due to small sample sizes. This makes the resulting data unstable and less reliable for local-level decision-making and resource allocation.

This project employs Small Area Estimation (SAE) techniques to overcome these limitations. By “borrowing strength” from auxiliary data and neighboring geographic areas, SAE models—specifically the Spatial Fay-Herriot and Unstratified Cluster-level models—aim to produce more precise, smoothed prevalence estimates. This study provides a comparative geospatial analysis of these models across Kenya’s 47 counties to identify the most robust method for informing localized public health interventions.

2 Methodology

2.1 Study Area

The study covers Kenya, focusing on the 47 counties (Level 2). This sub-national focus is critical because national averages often mask the severe disparities found in Arid and Semi-Arid Lands (ASAL) like Mandera

and Turkana compared to urban hubs like Nairobi or Kiambu.

2.2 Data Sources

The primary source is the **Kenya Demographic and Health Survey (KDHS)**.

2.2.1 Cases Data

i. Cases Data (Numerator): “Cases” refer to the number of women aged 15–49 who reported attending at least four antenatal care (ANC4+) visits during their most recent pregnancy.

2.2.2 Population Data

Population Data (Denominator): High-resolution population data is sourced from the gridded population estimates from WorldPop, which provide the total number of women of reproductive age per administrative unit.

2.2.3 Expected Cases

Expected Cases Data: This is a calculated value based on applying the national average rate of ANC4+ attendance to the specific sub-county population. It serves as a baseline to determine if a sub-county is performing “above” or “below” expectations given its size.

2.3 Statistical analysis

2.3.1 Descriptive and Exploratory Statistics

Direct Estimates: Calculating the weighted proportion of ANC4+ visits directly from KDHS data for each sub-county.

Variance Assessment: Calculating the Coefficient of Variation (CV) and standard deviation to identify areas where survey data is too thin (unreliable).

2.3.2 Spatial Modelling and Model fitting

2.3.2.1 Direct Estimates Direct estimates uses raw, weighted data collected from household surveys within each specific county to calculate prevalence rates. For y_j the binary outcome of interest for the j th individual in the survey and w_j the associated design weight, the weighted estimator for area i is

$$\hat{p}_i^W = \frac{\sum_{j \in S_i} y_j w_j}{\sum_{j \in S_i} w_j}$$

where S_i is the set of individual indices within the i -th region.

The confidence intervals are computed on the logit scale. Let D_i denote the design-based variance of $\hat{p}_i^W$. The asymptotic design-based variance of $\text{logit}(\hat{p}_i^W)$ is

$$V_i = \frac{D_i}{(\hat{p}_i^W)^2 (1 - \hat{p}_i^W)^2}.$$

The confidence interval on the probability scale is then computed as

$$\left(\text{expit} \left[\text{logit}(\hat{p}_i^W) - z_{\alpha/2} \sqrt{V_i} \right], \text{expit} \left[\text{logit}(\hat{p}_i^W) + z_{\alpha/2} \sqrt{V_i} \right] \right).$$

. This method does not incorporate any additional information beyond the survey data itself. As a result, direct estimates can be highly variable, especially in areas with small sample sizes or low event counts. The

estimates may be unstable and have wide confidence intervals, making them less reliable for decision-making at the local level.

2.3.2.2 Fay-Herriot Model Fay-Herriot Model is an area-level model that combines direct survey estimates with auxiliary information available at the area level (e.g., census data, administrative records). The model provides smoothed estimates at the areal level using the direct estimates $\hat{p}_i^W$ as input. The direct estimates are treated as noisy observations of the true underlying prevalence, with the noise variance given by the design-based variance. The model assumes that the true prevalence in each area is related to the auxiliary variables through a linear regression framework. It also incorporates random effects to account for unexplained variability between areas. By “borrowing strength” from related areas and covariates, the Fay-Herriot model produces smoothed estimates with reduced variance compared to direct estimates. This results in more stable and reliable prevalence estimates, particularly in areas with limited survey data. Consider a spatial Fay-Herriot model for the logit-transformed direct estimates, defined as

$$\text{logit}(\hat{p}_i^W) \mid \lambda_i \sim \text{Normal}(\lambda_i, V_i),$$

where V_i is the design-based variance of $\text{logit}(\hat{p}_i^W)$.

The latent process is specified as

$$\lambda_i = \alpha + e_i + S_i,$$

where α is an overall intercept, e_i represents unstructured random effects, and S_i represents structured spatial random effects. The latent true prevalence is given by $\text{expit}(\lambda_i)$.

Inference is carried out in a Bayesian framework by assigning prior distributions to α , e_i , and S_i , along with hyperpriors for their variance parameters.

2.3.2.3 Unstratified Cluster-level Model Unstratified Cluster-level Model is a unit-level model that utilizes individual survey data along with auxiliary information at the cluster or individual level. This model accounts for the hierarchical structure of the data and incorporates spatial correlation between neighboring areas. By modeling the data at a finer resolution, the Unstratified Cluster-level model can capture local variations in prevalence more effectively. It also “borrows strength” from both auxiliary variables and spatially correlated areas, leading to further reductions in variance and improved precision of the estimates. This model is particularly useful in settings with complex survey designs and spatial dependencies. The robustness of the Unstratified Cluster-level model often results in the most precise estimates among the three methods, making it a preferred choice for small area estimation in public health research. Let Y_x denote the number of events in cluster x , and let n_x be the number of individuals at risk, for $x = 1, \dots, X$. The unstratified model assumes the following hierarchical structure:

$$Y_x \mid p_x, d \sim \text{Beta-Binomial}(n_x, p_x, d),$$

$$p_x = \text{expit}(\alpha + e_{i[s_x]} + S_{i[s_x]}),$$

where α is the intercept and $i[s_x]$ indexes the area within which cluster s_x resides.

The random effects e_i and S_i represent unstructured and structured spatial effects, respectively, and are assigned the same priors as in the area-level model.

The Beta-Binomial distribution arises from a hierarchical formulation in which the cluster-specific probability follows a $\text{Beta}(a, b)$ prior. The overdispersion parameter

$$d = \frac{1}{a + b + 1},$$

takes values in $(0, 1)$ and represents the intraclass correlation between Bernoulli trials within a cluster. The default prior for the overdispersion parameter is

$$\text{logit}(d) \sim \mathcal{N}(0, 0.4).$$

3 Results and Discussion

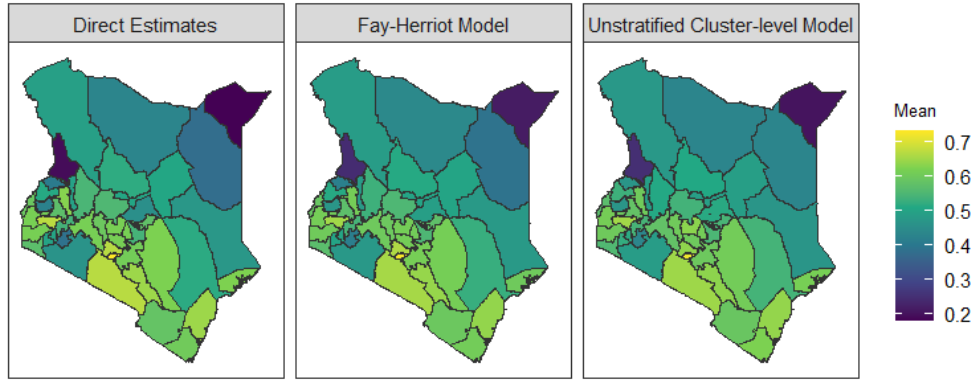


Figure 1: Maps comparing the Mean ANC4+ prevalence estimates from Direct Estimation, Fay-Herriot Model and Unstratified Cluster-level Model across Kenyan counties admin 1.

The maps 1 display the estimated ANC4+ (at least four antenatal care visits) prevalence across the counties of Kenya at the admin level 1, comparing a direct survey estimate with two Small Area Estimation (SAE) models: Spatial Fay-Herriot model and Unstratified Cluster-level model. All the three maps clearly indicate that ANC4+ utilization rates are suboptimal in parts of the country and that substantial subnational variation exists. The main difference is the smoothing effect that the models apply compared to the raw direct estimates. The maps consistently highlight significant geographical disparities in ANC4+ coverage. The colour scale ranges from 0.2 (20% prevalence) in dark purple to 0.7 (70% prevalence) in bright yellow/green. The highest prevalence rates (bright yellow/green shades) are concentrated in the central highlands (Kiambu, Nyeri, Murang'a), central counties like Nairobi, Nakuru and parts of the coast (Mombasa, Kilifi), western regions of Vihiga, indicating better access to antenatal care services in these regions. In contrast, the lowest prevalence rates (dark purple shades) are observed in the arid and semi-arid northern and eastern counties.

(Mandera, Wajir, Garissa, Marsabit, Turkana), reflecting challenges in healthcare access and utilization in these areas. The models provide more stable estimates in areas where direct estimates have been unreliable due to a lack of sufficient survey data.

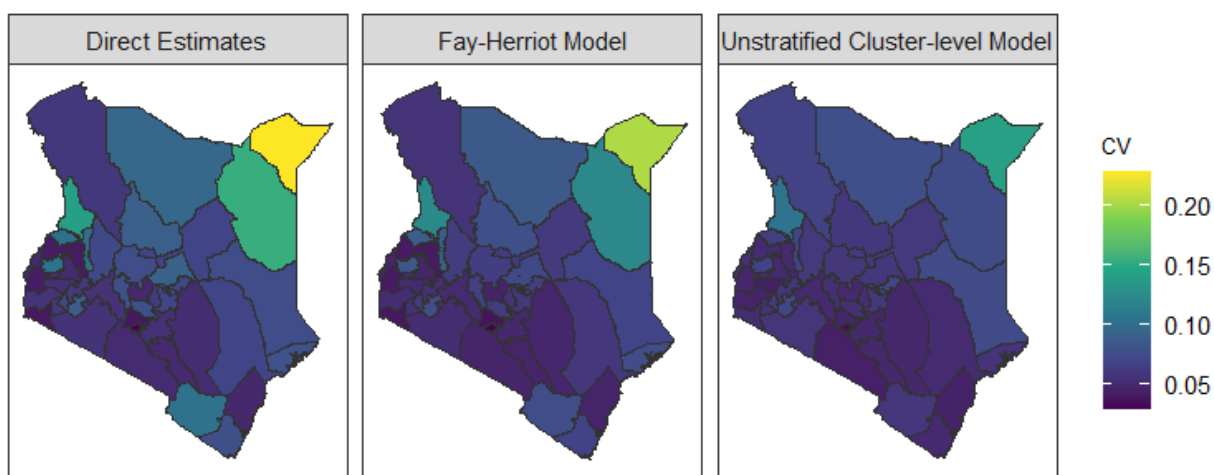


Figure 2: Maps comparing the Coefficient of Variation (CV) of ANC4+ prevalence estimates from Direct Estimation, Fay-Herriot Model and Unstratified Cluster-level Model across Kenyan counties admin 1.

The maps 2 display the Coefficient of Variation (CV) for health indicator estimates across Kenyan counties, demonstrating the reduction in uncertainty achieved by using Small Area Estimation (SAE) models. The color scale indicates higher CV (more uncertainty) with brighter yellow/green (up to 0.20 or 20%) and lower CV (less uncertainty/more precise estimates) with dark purple (down to 0.05 or 5%). Direct estimates map shows the raw uncertainty from survey data. highlighting areas where data is sparse resulting in the highest CV values. High CV (yellow/green) is prominent in northern and eastern counties (Mandera, Wajir, Garissa, Marsabit, Turkana) where populations are sparse and survey data is limited. Fay-Herriot model introduces smoothing which reduces the uncertainty by “borrowing strength” from auxillary data and neighbouring areas. This map is significantly darker overall. The high CV hotspots seen in the north/east and west have

been largely reduced to mid-level uncertainty (blue). Unstratified cluster-level model best accounts for spatial correlation. The map is the darkest (dark purple below 0.10 CV) showing the lowest CV across the board. It achieves the most robust and stable standard errors across all regions.

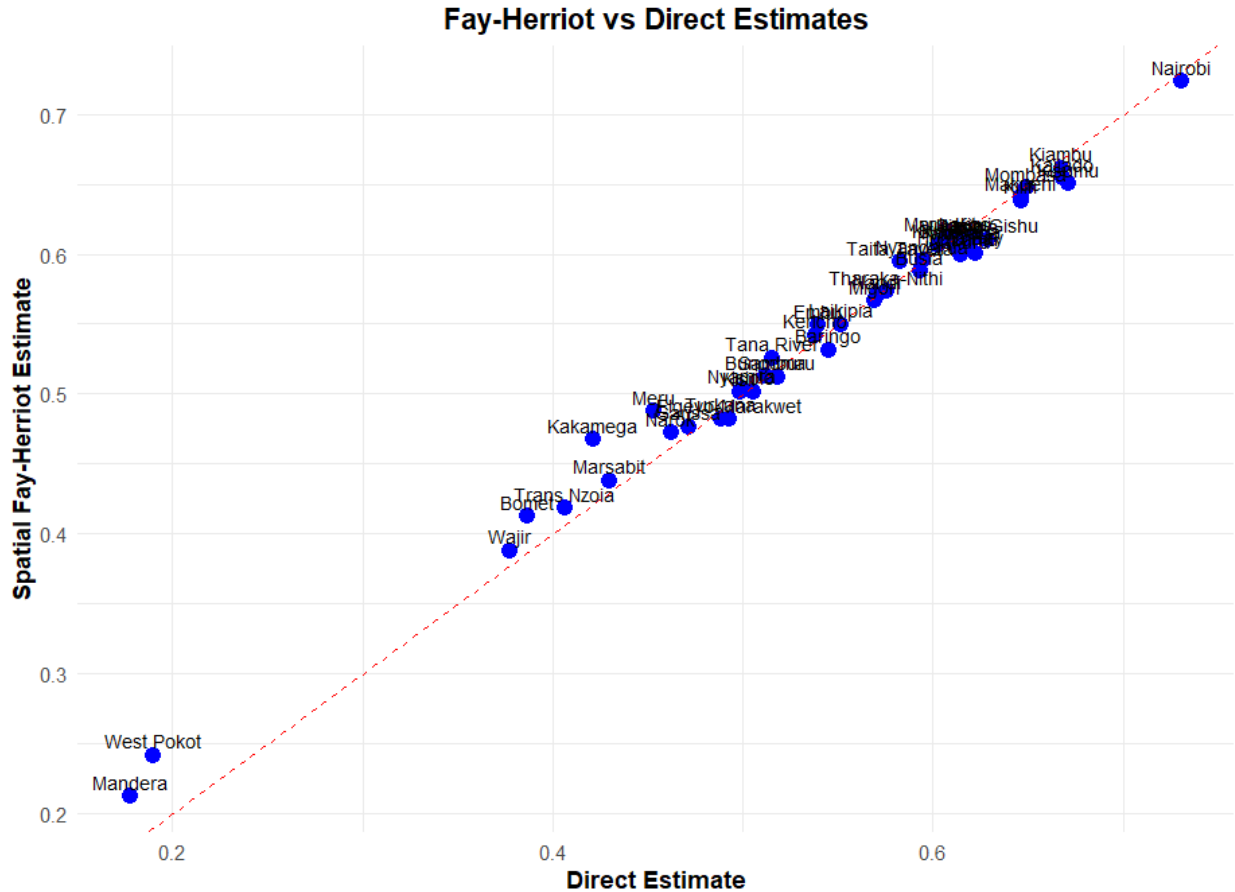


Figure 3: Scatter plot comparing ANC4+ prevalence estimates from Direct Estimation and Fay-Herriot Model across Kenyan counties.

There is a strong positive linear relationship between spatial fay-herriot and direct estimation model. This strong alignment suggests the Fay-Herriot model is working well, as it is utilizing reliable covariate information that match the patterns in the direct survey data. Areas that sit tightly on the line of equality benefit from high-quality, relevant covariates and likely have sufficient sample sizes in the original survey data so the model doesn't need to adjust them much. Nairobi is highly urbanized with high prevalence approximately 0.7 sitting almost perfectly on the line. census demographic, wealth index and access to facilities data accurately reflect the survey results for the coastal and central (Mombasa, nakuru, kiambu) areas clustering tightly on the line with a prevalence mark of 0.5. These western counties (kakamega, bungoma) show Minor Shrinkage. Northern counties (mandera, west pokot) are remote and sparsely populated with the lowest prevalence 0.2. The direct estimates are less stable on their own 3.

The comparison of Standard Errors (SE) between Direct Estimation and a Spatial Fay-Herriot Model demonstrating the model's ability to reduce uncertainty 4. All data points lie conclusively below the dashed line of equality. For every area in the plot, the standard error from the Fay-Herriot Model is lower than that from the Direct Estimation method. Areas with the highest initial uncertainty (Wajir, Taita Taveta, Lamu, Elgeyo-Marakwet) show the most substantial reduction in their standard errors when modeled using the Fay-Herriot approach. Areas with low initial uncertainty, like Mombasa, Kilifi, and Bungoma (bottom left), show less improvement but still sit below the line. Spatial Fay-Herriot Model provides more precise

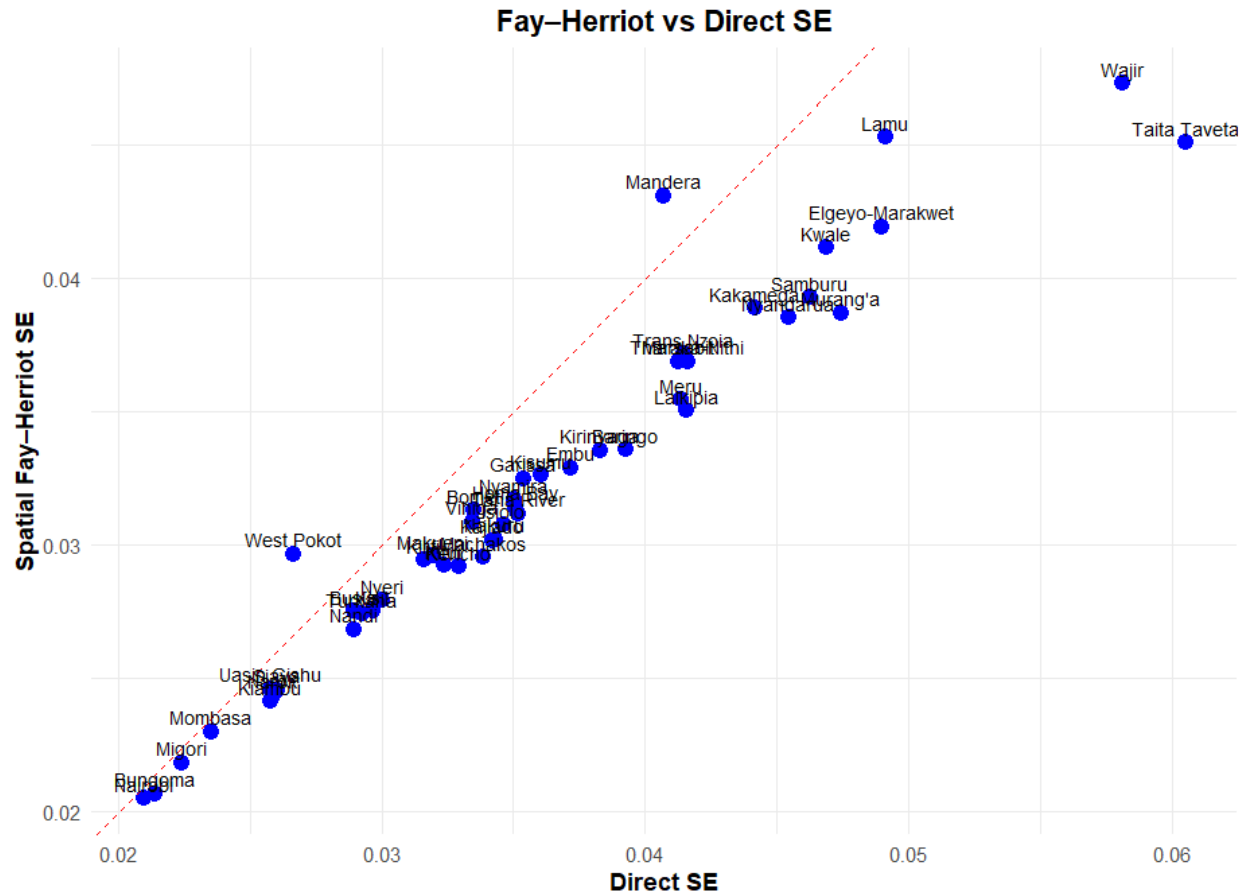


Figure 4: Scatter plot comparing standard errors (SE) of ANC4+ prevalence estimates from Direct Estimation and Fay-Herriot Model across Kenyan counties.

and stable estimates across all regions by reducing the uncertainty inherent in raw survey data, making the overall inference more reliable for local decision-making.

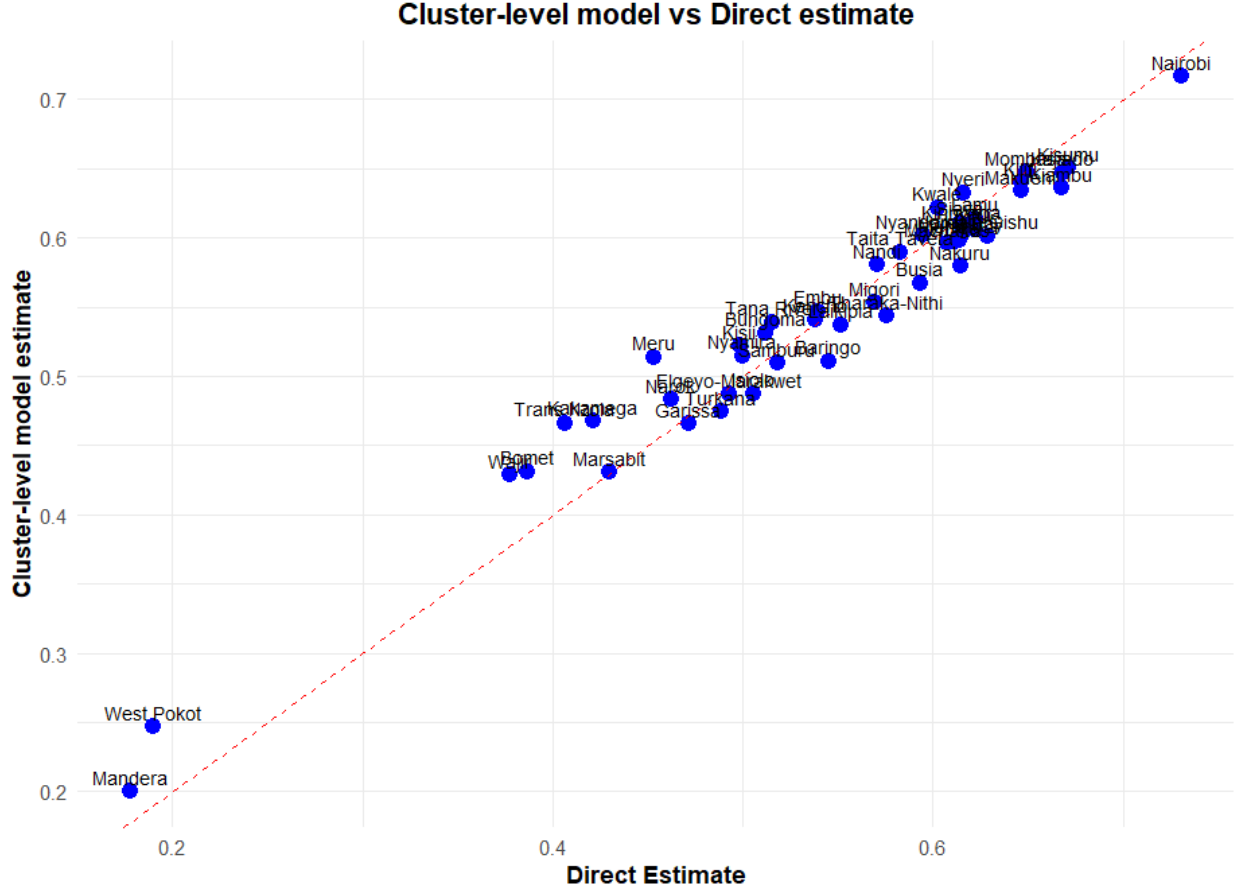


Figure 5: Scatter plot comparing ANC4+ prevalence estimates from Direct Estimation and Unstratified Cluster-level Model across Kenyan counties.

Most counties cluster around the 0.4 to 0.6 range, both methods yield very similar estimates as the points sit close to the diagonal line. Differences appear in areas with extreme values, counties like Mander and West Pokot have the lowest estimates 0.2. the cluster-level estimates for these counties are slightly higher closer to 0.22-0.25 pulling them towards the overall average. Nairobi has the highest direct estimate 0.7. Its cluster-level estimate is slightly lower pulling it down towards the average. This is due to statistical smoothing effect where the cluster-level model borrow information from neighbouring areas and the overall population average. This pulls the extreme direct estimates closer to the mean especially when the direct estimates might be unreliable due to small sizes or high variability.

The plot 6 demonstrates a strong positive linear relationship between Unstratified cluster level model and direct estimation methods, areas with higher direct estimation SEs also have higher cluster-level SEs but the cluster-level model consistently shrinks the SEs downward. The Unstratified Cluster-level Model consistently reduces the standard errors of the estimates compared to using raw survey data alone. For every county in the plot, the standard error from the Cluster-level Model is lower than that from the Direct Estimation method. showing the “borrowing strength” principle of small area estimation techniques. Most counties show a typical reduction in SEs, clustering along the primary trend line below the dashed line a result of standard covariates (like urban/rural status, wealth, education) effectively stabilizing the estimates. Nairobi, Mombasa, Kilifi, Bungoma, and counties in the central highlands (Nyeri, Embu, Kirinyaga) all show lower SEs with the model and follow the general trend. Deviations occur in areas with extremely high initial

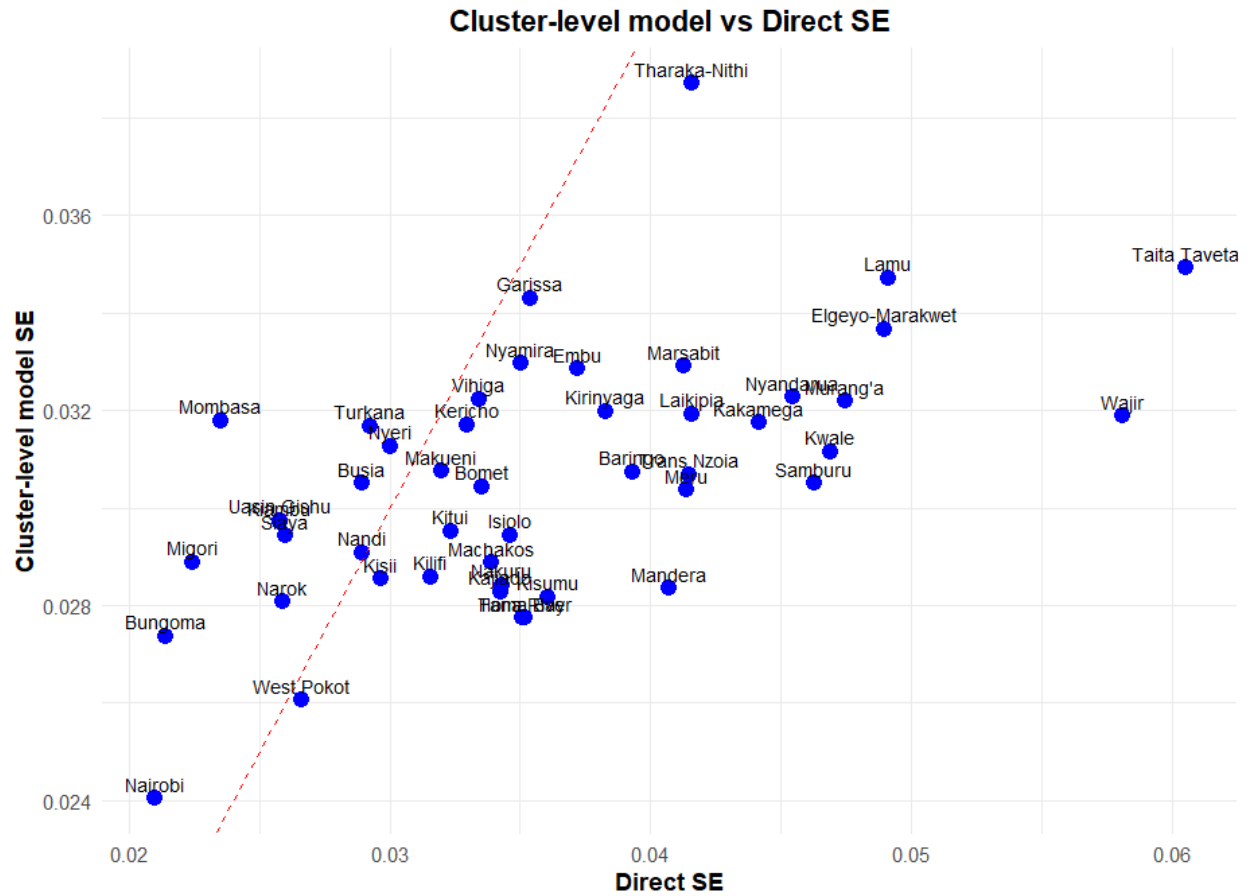
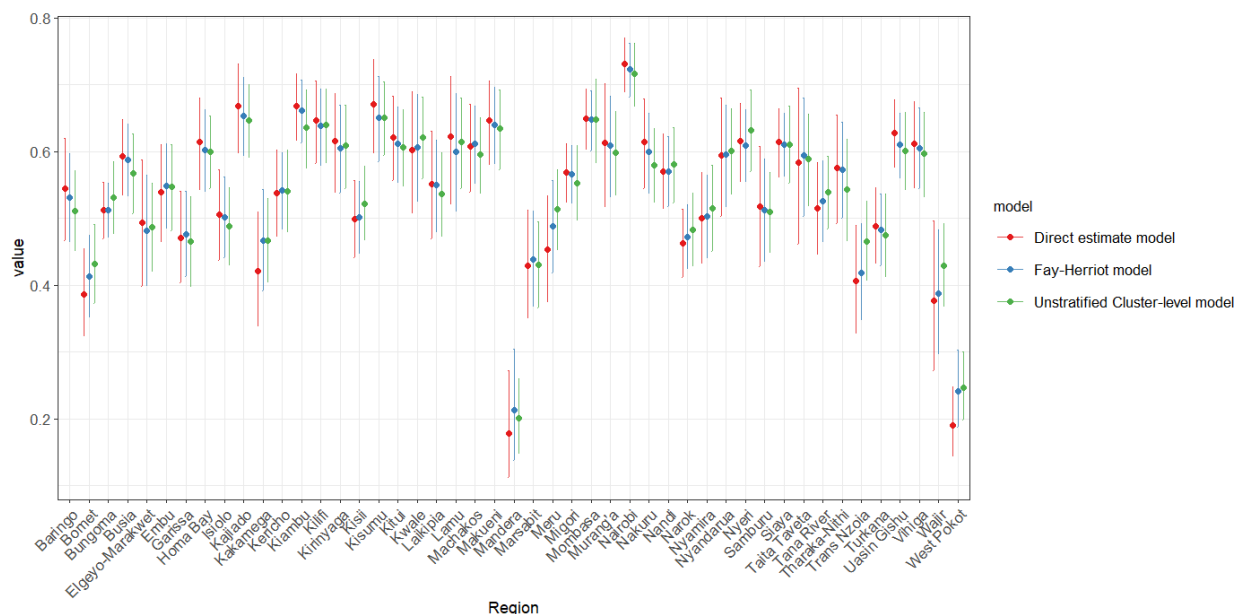
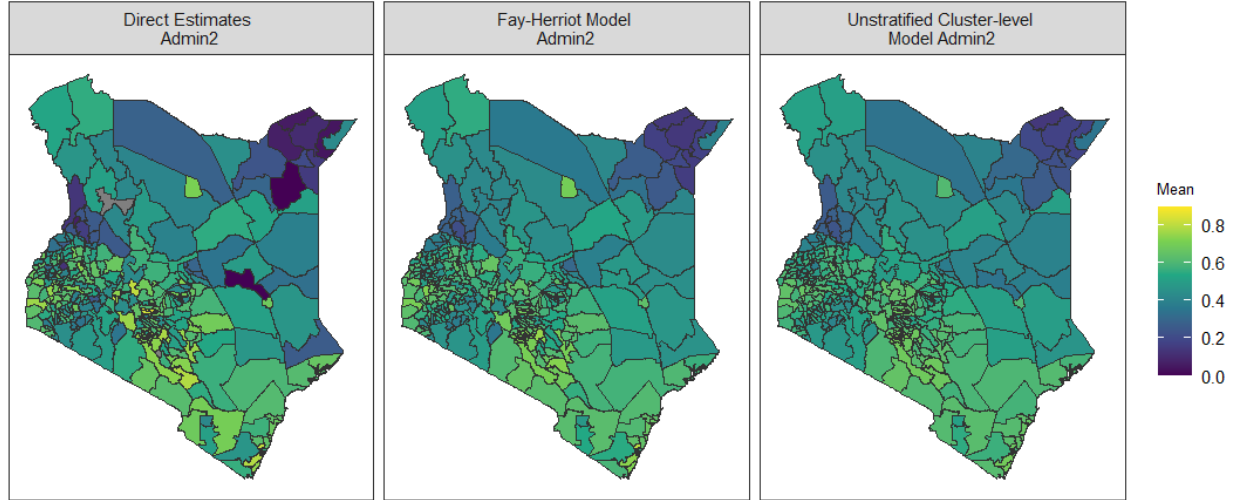


Figure 6: Scatter plot comparing standard errors (SE) of ANC4+ prevalence estimates from Direct Estimation and Unstratified Cluster-level Model across Kenyan counties.

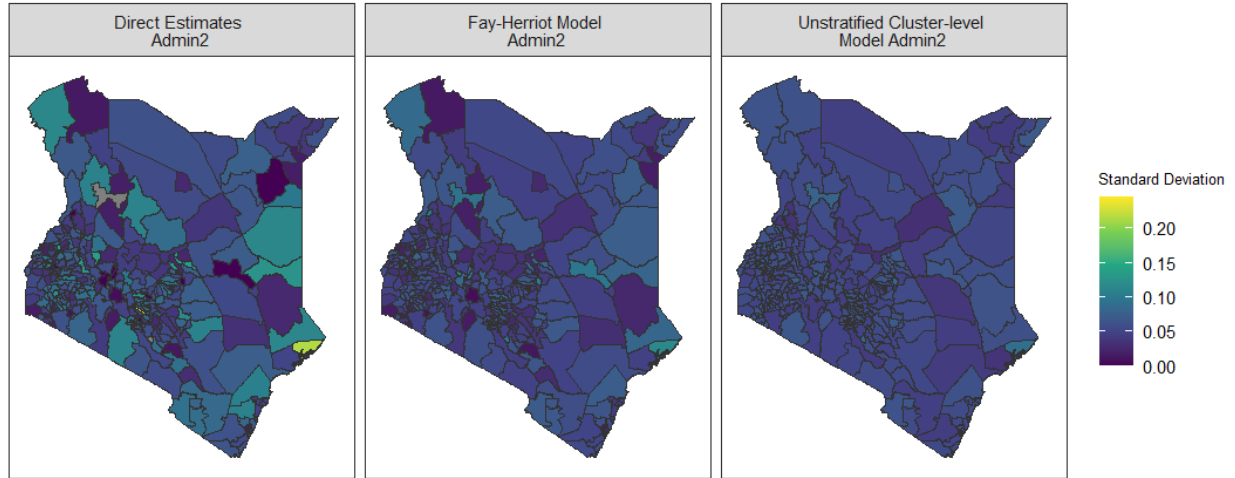
uncertainty in the raw survey data. These points show the largest vertical drop when modeled, meaning they receive the most benefit from the spatial modeling technique. Wajir, Taita Taveta, Lamu: These counties exhibit the highest direct SEs (around 0.05 to 0.06). Their large drop in SE (down to about 0.03 to 0.035) highlights that the model significantly improved precision in these areas, likely due to low sample sizes in the original surveys. Tharaka-Nithi county shows a substantial reduction in SE, indicating a strong benefit from model-based precision improvements. The Unstratified Cluster-level Model provides more reliable standard errors across all regions, particularly in remote or sparsely populated areas, by reducing the uncertainty inherent in raw survey data. This makes the resulting confidence intervals tighter and the overall inference more reliable leading to better-informed local health decisions.



The indicator prevalence ranges significantly, with most estimates falling between 0.4 (40%) and 0.7 (70%) ???. The lowest estimates are near 0.2 (20%) in counties like Mandera and Wajir. For many counties in the middle range (Bungoma, Kakamega Kisumu), the three estimates cluster closely together, suggesting high agreement when data is sufficient. The red (direct estimate) error bars are visually the longest, indicating the highest uncertainty in the raw data, especially in rural or sparsely populated areas. The blue and green error bars (SAE models) are significantly shorter demonstrating the “borrowing strength” effect, where the models reduce standard errors and increase precision. In areas like Mandera and Wajir, outlier areas, the direct estimates (red dot) have very wide error bars, but the model estimates (blue/green dots) are more stable and precise (shorter bars). The models also “shrink” the estimates slightly toward the mean in these sparse areas, as seen by the blue and green dots being slightly different from the red dot. The plot highlights the effectiveness of small area estimation (SAE) models in producing more stable and reliable health indicator estimates across all regions, particularly in those counties where traditional survey methods yield high uncertainty.



The geographic variation of antenatal care utilization (ANC4+) at the sub-county (Admin 2) level in Kenya using different estimation techniques range from low prevalence, ~ 0.0 to high prevalence, ~ 0.8 . ANC4+ coverage is highly uneven across Kenya ???. Higher utilization is generally concentrated in central and western regions, while the lowest rates are found in northern and northeastern areas, such as Mandera West, which has recorded prevalence as low as 17%. Direct Estimates rely solely on survey data from each area. They are “noisy” and show gaps (grey areas) where sample sizes are too small to provide a reliable estimate at the sub-county level. Fay-Herriot Model “borrows strength” from neighboring areas to provide smoother and more precise estimates for sub-counties with limited data. Unstratified Cluster-level model further refines the spatial distribution by leveraging cluster-level data to map disparities more granularly across the 295 sub-counties. While Kenya’s national average for ANC4+ utilization is approximately 60.8%, none of the three countries in this regional study (Kenya, Uganda, Tanzania) had met the global 2025 target of 70% coverage at the national level by the survey years. Disparities in these plots are often tied to wealth, education, and geographic accessibility. Women in wealthier quintiles or with higher education levels are significantly more likely to reach the ANC4+ threshold



The transition from left to right demonstrates the power of Small Area Estimation (SAE). While direct survey data is often too “noisy” to guide local policy in every sub-county, model-based approaches successfully minimize the standard deviation, providing the reliable, high-resolution data needed to target health interventions in underserved areas like Mendera West or West Pokot ???. Direct Estimates Admin 2 map shows the highest and most widespread uncertainty. Many areas exhibit high SD (green to yellow), particularly in northern and sparsely populated regions. This is common with direct estimates because they rely only on survey data from that specific sub-county. If the sample size is small, the resulting estimate is often unreliable and has high variance. The grey patches which represent areas where no data was available to form a direct estimate. Fay-Herriot Model Admin 2 there is a significant reduction in uncertainty compared to direct estimates. The map is predominantly dark purple, indicating much lower SD values. This model “borrows strength” by using auxiliary variables and data from neighboring areas to stabilize estimates in regions with small sample sizes. Unstratified Cluster-level Model Admin 2 model shows the lowest overall uncertainty (most precise estimates). By leveraging cluster-level data directly rather than just area-level averages, this model further smooths the variance across the 295 sub-counties. It provides a more uniform, high-precision map, which is critical for identifying true geographic inequalities in antenatal care access.